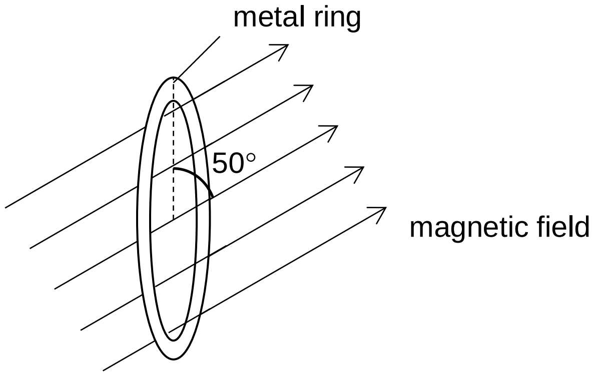
start from zero.

**26** A uniform magnetic field directed at 50° from the vertical passes through a circular metal ring of diameter 0.50 m and resistance 3.0 Ω.




The magnetic flux density through the ring decreases by $4.0 \times 10^{-5}$ T at a constant rate in 2.0 s. During this change, the current induced in the ring

**A** remains constant at 1.0 $\mu$A.

**B** remains constant at 1.3 $\mu$A.

**C** decreases from zero to 1.0 $\mu$A.

**D** decreases from zero to 1.3 $\mu$A.

Ans: A

$$I = \frac{\varepsilon}{R} = \frac{\Delta BA}{\Delta t}\left(\frac{1}{R}\right) = \frac{(4.0 \times 10^{-5})\sin 50(\pi(\frac{0.50}{2})^2)}{2.0}\left(\frac{1}{3.0}\right) = 1.0 \times 10^{-6}\text{ A}$$

The current remains constant as the rate of change of flux is constant.

**27** When a light bulb is connected across an a.c. source of peak voltage 170 V, the power